SOLOBOLO / RIEMANN-HYPOTHESIS-FINAL

Hilbert-Vector Resolvents and the Riemann Hypothesis

A detailed analytic seed criterion

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Abstract

We give a precise, self-contained analytic formulation of a Hilbert-vector anchor for the Riemann hypothesis. Let Θ be the entire function determined by $\Theta(w^2) = \xi(\frac{1}{2} + w)$, and put $L(a) = \Theta'(a)/\Theta(a)$ for $a \geq 0$. We prove that RH is equivalent to the existence of one Hilbert space, one bounded nonnegative self-adjoint operator K, and one fixed vector v such that $\frac{\mathrm{d}}{\mathrm{d}a}(aL(a)) = \|(I + aK)^{-1}v\|^2$ on a nonempty positive interval. The derivation includes the resolvent calculus, its divided-difference Gram kernel, and the analytic continuation that excludes zeros off the negative real axis. A possible C/a ambiguity is removed using $\Theta(0) \neq 0$, rather than suppressed by an unstated normalization. The reverse implication constructs a positive diagonal operator from the zero product under RH, with multiplicities counted correctly. We also develop the spectral-measure and all-order moment formulations, explain basepoint shifts, and prove that no exactly calibrated finite-dimensional model can suffice. The result is an analytic equivalence; it does not construct the required seed unconditionally or produce an accepted finite witness. Hyperbolic zeta identities additionally give explicit comparison models and a conditional determinant-limit route to the actual seed.

Evidence and scope. Every proof in this note is an on-paper derivation. No Lean compiler or axiom audit was run. The seed predicates below are defined explicitly here; they are not asserted to be recovered field-for-field definitions of a legacy `HasSeed` declaration. The repository's finite-checker target remains open. Classical background is cited, and no novelty or unconditional proof of RH is claimed.

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1 The exact assertion and its role

The starting point is the family of identities

$$R_a = (I + aK)^{-1}, \qquad \frac{\mathrm{d}}{\mathrm{d}a}(aL(a)) = \|R_a v\|^2, \qquad V_a = 2\sqrt{a}\, R_a v. \tag{1.1}$$

Their significance depends on their quantifiers. The operator and vector must be the *same* for every parameter in the interval, and the scalar L must come from the actual completed Riemann zeta function. An arbitrary nonnegative scalar function can be represented as a squared norm by allowing its vector to vary freely. Such a pointwise representation contains none of the resolvent compatibility used below.

We fix the standard completion

$$\xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \tag{1.2}$$

with the removable values understood. The symmetry $\xi(s) = \xi(1-s)$ and normalization are classical [1]. Section 2 defines Θ without choosing a square-root branch and proves that L is a well-defined real analytic function on $[0, \infty)$.

Definition 1.1 (Calibrated Hilbert seed)**.** The assertion HSeed_Θ means that there exist a complex Hilbert space $\mathcal{H}$, an operator $K \in \mathcal{B}(\mathcal{H})$, a vector $v \in \mathcal{H}$, and real numbers $0 < \alpha < \beta < \infty$ such that

$$K = K^*, \qquad \langle x, Kx\rangle \geq 0 \quad (x \in \mathcal{H}), \tag{1.3}$$

and, for every $a \in (\alpha, \beta)$,

$$L(a) = \langle v, (I + aK)^{-1} v\rangle. \tag{1.4}$$

The inner product is conjugate-linear in its first slot and linear in its second slot. In particular, the complex extension $z \mapsto \langle v, (I + zK)^{-1}v\rangle$ is holomorphic where the inverse exists.

Definition 1.2 (Differential Hilbert seed)**.** The assertion DSeed_Θ has the same existential data and operator hypotheses as Definition 1.1, but replaces (1.4) by

$$\frac{\mathrm{d}}{\mathrm{d}a}(aL(a)) = \|(I + aK)^{-1}v\|^2 \quad \text{for every } a \in (\alpha, \beta). \tag{1.5}$$

This is an exact functional identity on an interval, not a test at finitely many sample points.

Theorem 1.3 (Main analytic criterion)**.** *For the actual Θ associated with (1.2),*

$$\mathrm{RH} \iff \mathsf{HSeed}_\Theta \iff \mathsf{DSeed}_\Theta. \tag{1.6}$$

If either seed identity holds on one nonempty positive interval, its calibration extends to every $a \geq 0$ for the same K and v. Under RH, a seed can be chosen on $\ell^2(\mathbb{N}; \mathbb{C})$ with K positive, compact and trace class. In that realization,

$$\operatorname{Tr} K = \|v\|^2 = \frac{\xi''(1/2)}{2\xi(1/2)}. \tag{1.7}$$

Sections 3–6 prove the forward and differential implications without assuming RH or a positive zero product. Section 7 proves the converse under RH. The theorem is a criterion: the existential seed assertion carries the mathematical difficulty of RH. No independently constructed seed is supplied here.

There are two different questions behind this note. The analytic question asks exactly what a fixed positive resolvent identity implies. The finite question asks whether a terminating checker on finite raw data can certify such an identity, and whether RH guarantees an accepted input to that same checker. The second question does not follow merely by naming the first one a seed. Section 11 makes this distinction precise.

2 The actual entire function and its zero geometry

2.1 Square descent without a branch choice

Put $E(w) = \xi(\frac{1}{2} + w)$. This is entire, and the functional equation makes it even. Its Taylor series has the form

$$E(w) = \sum_{n=0}^{\infty} c_n w^{2n}, \qquad c_n = \frac{\xi^{(2n)}(1/2)}{(2n)!}. \tag{2.1}$$

Define

$$\Theta(z) = \sum_{n=0}^{\infty} c_n z^n. \tag{2.2}$$

The series has infinite radius of convergence. Indeed, for any $r > 0$, the absolute convergence of (2.1) at a point with $|w| = \sqrt{r}$ gives $\sum_n |c_n| r^n < \infty$. Consequently

$$\Theta(w^2) = \xi(\tfrac{1}{2} + w) \qquad (w \in \mathbb{C}). \tag{2.3}$$

The shorthand $\Theta(z) = \xi(\frac{1}{2} + \sqrt{z})$ therefore has a single-valued meaning: the two square roots give the same value. The branch cut introduced later belongs to the resolvent domain, not to Θ.

Complex conjugation preserves ξ, as follows first on its defining real domain and then by analytic continuation. Thus the coefficients c_n are real, and $\Theta(\overline{z}) = \overline{\Theta(z)}$.

2.2 Positivity on the real positive axis

For completeness, the nonvanishing needed to define L can be checked without RH. For $0 < s < 1$, the alternating-series formula [2, Eq. 25.2.3] gives

$$\zeta(s) = \frac{\eta(s)}{1 - 2^{1-s}}, \qquad \eta(s) = \sum_{n=1}^{\infty} (-1)^{n-1} n^{-s} > 0. \tag{2.4}$$

The strict inequality follows by grouping consecutive terms: $(2j-1)^{-s} - (2j)^{-s} > 0$. The even partial sums converge to the same alternating-series limit and already their first pair is positive. The denominator in (2.4) is negative, so $\zeta(s) < 0$. Since $s(s-1) < 0$ and the remaining factors in (1.2) are positive, $\xi(s) > 0$ on $(0, 1)$. For $s > 1$, every factor in (1.2) is positive, and the removable value at $s = 1$ is $1/2$. Applying this with $s = 1/2 + \sqrt{a}$ proves

$$\Theta(a) > 0 \quad (a \geq 0), \qquad \Theta(0) = \xi(1/2) > 0. \tag{2.5}$$

It follows that

$$L(a) = \frac{\Theta'(a)}{\Theta(a)} \in \mathbb{R} \qquad (a \geq 0) \tag{2.6}$$

is real analytic, including at zero. On this axis it equals the recovered quantity $\mathrm{Re}(\Theta'(a)/\Theta(a))$; taking the real part loses no information there. It would not be legitimate to make that replacement for arbitrary complex z.

Expanding at zero gives the exact normalization

$$\kappa := L(0) = \frac{\Theta'(0)}{\Theta(0)} = \frac{\xi''(1/2)}{2\xi(1/2)}. \tag{2.7}$$

No numerical approximation to κ is needed in any proof below.

2.3 The critical line becomes the negative real axis

The zeros of ξ are exactly the nontrivial zeros of ζ; the standard zero symmetry and the meaning of RH are recalled in [3]. If $\rho = 1/2 + w$ is such a zero, then $z = w^2$ is a zero of Θ, and conversely every zero of Θ arises this way by (2.3).

For $w = x + iy$,

$$w^2 = (x^2 - y^2) + 2ixy. \tag{2.8}$$

If w^2 is real and nonpositive, then $xy = 0$ and $x^2 - y^2 \le 0$. If $y = 0$, these imply $x = 0$; if $y \neq 0$, the equality $xy = 0$ again implies $x = 0$. Conversely $x = 0$ gives $w^2 = -y^2 \le 0$. Since $\Theta(0) \neq 0$, its zeros never correspond to $w = 0$. We have therefore proved

Proposition 2.1 (Exact zero geometry)**.**

$$\mathrm{RH} \quad \Longleftrightarrow \quad \big(\Theta(z) = 0 \implies z \in (-\infty, 0)\big). \tag{2.9}$$

This elementary change of variables is the final geometric step of the forward proof. It refers to the actual ξ and Θ throughout.

3 Resolvent calculus with one fixed vector

Fix a complex Hilbert space $\mathcal{H}$, a bounded nonnegative self-adjoint operator K, and a vector v. For $a \ge 0$, set

$$R_a = (I + aK)^{-1}, \qquad f(a) = \langle v, R_a v\rangle, \qquad g(a) = af(a). \tag{3.1}$$

At this stage f is a model scalar function; equality with the actual L has not been assumed.

3.1 Invertibility, self-adjointness and commutation

For every $x \in \mathcal{H}$, positivity and Cauchy–Schwarz imply

$$\|x\|^2 \le \langle x, (I + aK)x\rangle \le \|x\|\,\|(I + aK)x\|.$$

Thus $I + aK$ is bounded below by 1. Its range is closed. Its orthogonal complement is the kernel of its adjoint, which equals its own kernel because the operator is self-adjoint. The lower bound makes that kernel zero, so the range is also dense and hence is all of $\mathcal{H}$. It follows that

$$\|R_a\| \le 1, \qquad R_a^* = R_a, \qquad R_a \ge 0. \tag{3.2}$$

For the last inequality write $y = (I + aK)x$ and observe that $\langle y, R_a y\rangle = \langle (I + aK)x, x\rangle \ge 0$. The operator K commutes with $I + aK$ and therefore with its inverse. The operators R_a and R_b also commute.

Lemma 3.1 (Two resolvent identities)**.** *For $a, b \ge 0$,*

$$R_a - R_b = (b - a)R_a K R_b, \tag{3.3}$$

$$aR_a - bR_b = (a - b)R_a R_b. \tag{3.4}$$

Proof. For the first identity, multiply out $R_a[(I + bK) - (I + aK)]R_b$. For the second, the same calculation with $a(I + bK) - b(I + aK) = (a - b)I$ gives the result. Both formulas also hold when $a = b$, with both sides then zero. □

3.2 Differentiation in operator norm

From (3.3),

$$\frac{R_{a+h} - R_a}{h} = -R_{a+h} K R_a.$$

Continuity of inversion, or the same identity and local boundedness, shows that the right side tends in operator norm to $-R_a K R_a$. Since K commutes with R_a,

$$R_a' = -K R_a^2. \tag{3.5}$$

Repeated differentiation yields

$$R_a^{(n)} = (-1)^n n! K^n R_a^{n+1} \qquad (n \geq 0). \tag{3.6}$$

One can verify the induction directly: differentiating R_a^{n+1} produces $n+1$ identical commuting terms $-K R_a^{n+2}$. Applying the continuous linear functional $T \mapsto \langle v, Tv \rangle$ gives

$$(-1)^n f^{(n)}(a) = n! \langle v, K^n R_a^{n+1} v \rangle \geq 0. \tag{3.7}$$

The last operator is positive, for example by the bounded self-adjoint functional calculus [6]. Thus f is completely monotone. This consequence alone will not characterize a calibrated seed.

3.3 The norm-square identity

The identity $(I + aK)R_a = I$ implies $R_a = R_a^2 + aK R_a^2$. Consequently

$$\begin{aligned} g'(a) &= f(a) + a f'(a) \\ &= \langle v, (R_a - aK R_a^2) v \rangle \\ &= \langle v, R_a^2 v \rangle = \|R_a v\|^2. \end{aligned} \tag{3.8}$$

This proves the central algebraic formula, including its exact coefficient. It involves $af(a)$, not $f(a)$ alone and not an unspecified change of basepoint. Differentiating further gives

$$(-1)^n g^{(n+1)}(a) = (n+1)! \langle v, K^n R_a^{n+2} v \rangle \geq 0 \qquad (n \geq 0). \tag{3.9}$$

In particular g is increasing and concave on the positive axis. If $v \neq 0$, then $R_a v \neq 0$ and $g'(a) > 0$.

4 The Gram kernel and its exact normalization

Define $V_a = 2\sqrt{a}\, R_a v$ for $a > 0$. Taking the scalar product of (3.4) with v and using self-adjointness gives, for $a \neq b$,

$$\frac{g(a) - g(b)}{a - b} = \langle v, R_a R_b v \rangle = \langle R_a v, R_b v \rangle. \tag{4.1}$$

Hence

$$\langle V_a, V_b \rangle = 4\sqrt{ab}\, \frac{a f(a) - b f(b)}{a - b} \qquad (a, b > 0,\ a \neq b). \tag{4.2}$$

If the calibration $f = L$ holds, this is precisely the recovered Hilbert-vector Gram identity.

Proposition 4.1 (Diagonal and positivity)**.** *The divided-difference kernel*

$$\mathcal{K}_g(a,b) = \begin{cases} \dfrac{g(a)-g(b)}{a-b}, & a \neq b, \\ g'(a), & a = b \end{cases} \tag{4.3}$$

is positive semidefinite. More precisely, for any finite collection $a_1, \dots, a_m > 0$ and complex numbers $c_1, \dots, c_m$,

$$\sum_{i,j=1}^{m} \overline{c_i} c_j \mathcal{K}_g(a_i, a_j) = \|\sum_{j=1}^{m} c_j R_{a_j} v\|^2 \geq 0. \tag{4.4}$$

The diagonal extension of (4.2) is

$$\|V_a\|^2 = 4a\, g'(a) = 4a\, \|R_a v\|^2. \tag{4.5}$$

Proof. Equation (4.1) gives all off-diagonal entries, while (3.8) gives the diagonal. Expanding the displayed squared norm proves (4.4); multiplication by the factors $2\sqrt{a_i}$ gives the scaled Gram matrix. □

For two distinct parameters, the determinant condition is the concrete inequality

$$\left|\frac{g(a)-g(b)}{a-b}\right|^2 \leq g'(a)g'(b). \tag{4.6}$$

It is a necessary condition tying different points together. Positivity of g' at every individual point does not imply it.

Example 4.2 (Even strong scalar positivity is insufficient)**.** Let $\widetilde{g}(a) = 1 - e^{-a}$ and $\widetilde{f}(a) = (1-e^{-a})/a$ for $a > 0$. Then $\widetilde{g}' > 0$, $\widetilde{g}'' < 0$, and

$$\widetilde{f}(a) = \int_0^1 e^{-at}\,\mathrm{d}t, \qquad (-1)^n \widetilde{f}^{(n)}(a) = \int_0^1 t^n e^{-at}\,\mathrm{d}t > 0.$$

Nevertheless, writing $d = a - b \neq 0$, the off-diagonal divided difference is

$$\mathcal{K}_{\widetilde{g}}(a,b) = e^{-(a+b)/2}\frac{\sinh(d/2)}{d/2}.$$

The determinant of the corresponding two-by-two matrix equals

$$e^{-a-b}\left[1 - \left(\frac{\sinh(d/2)}{d/2}\right)^2\right] < 0. \tag{4.7}$$

The strict inequality follows from $|\sinh x| > |x|$ for nonzero real x, which follows by integrating $\cosh t > 1$ away from zero. Thus these scalar monotonicity conditions cannot be substituted for the fixed-resolvent structure.

5 Holomorphic extension of the model

Choose a real number $M > \|K\|$ with $M > 0$, and set

$$D_M = \mathbb{C} \setminus (-\infty, -1/M]. \tag{5.1}$$

This open connected domain contains zero and the entire positive real axis. The bounded self-adjoint spectral theorem puts $\sigma(K)$ in $[0, M]$ [6]. For $z \in D_M$ and $t \in [0, M]$, the number $1 + zt$ is nonzero. Indeed, a zero would require $t > 0$ and $z = -1/t \leq -1/M$. The functional calculus therefore defines

$$R_z = (I + zK)^{-1}, \qquad F(z) = \langle v, R_z v\rangle \qquad (z \in D_M). \tag{5.2}$$

Lemma 5.1 (Norm-holomorphic resolvent)**.** *The maps $z \mapsto R_z$ and $z \mapsto F(z)$ are holomorphic on D_M.*

Proof. Fix $z_0 \in D_M$. For sufficiently small h, $\|hKR_{z_0}\| < 1$ and

$$I + (z_0 + h)K = (I + z_0K)(I + hKR_{z_0}),$$
$$R_{z_0+h} = R_{z_0} \sum_{n=0}^{\infty} (-hKR_{z_0})^n.$$

The series converges in operator norm on a neighborhood of $h = 0$, so it is a norm-convergent local power series. The scalar functional $T \mapsto \langle v, Tv\rangle$ is continuous and complex linear under the convention in Definition 1.1, and applying it proves the scalar assertion. □

It is useful to retain zero in D_M. A cut at $(-\infty, 0]$ would still suffice for the normalized seed-to-RH argument, but it would hide the evaluation at zero that removes the integration constant in the differential seed. Boundedness of K supplies a genuine neighborhood of zero on which the resolvent is analytic.

The spectral theorem also gives a scalar measure. If E_K is the projection-valued spectral measure, define

$$\mu(B) = \langle v, E_K(B)v\rangle \qquad (B \text{ Borel in } [0, M]). \tag{5.3}$$

It is a finite positive measure, with total mass $\|v\|^2$, and

$$F(z) = \int_{[0,M]} \frac{1}{1 + zt}\, \mathrm{d}\mu(t). \tag{5.4}$$

For a compact $C \subset D_M$, the continuous function $|1 + zt|$ has a strictly positive minimum on $C \times [0, M]$. This bounds every fixed derivative of the integrand by an integrable constant. Thus

$$F^{(n)}(z) = (-1)^n n! \int_{[0,M]} \frac{t^n}{(1 + zt)^{n+1}}\, \mathrm{d}\mu(t). \tag{5.5}$$

This justifies the integral differentiations directly and independently checks the operator formulas of Section 3.

6 The analytic transfer to RH

6.1 Exact calibration on an interval is enough

Theorem 6.1 (Normalized seed implies RH)**.** *If HSeed_Θ holds, then* RH *holds. For its fixed seed data, $F(a) = L(a)$ for all $a \geq 0$.*

Proof. On the interval in Definition 1.1, the calibration and $\Theta(a) > 0$ imply

$$\Theta'(a) - F(a)\Theta(a) = 0.$$

The function

$$G(z) = \Theta'(z) - F(z)\Theta(z) \tag{6.1}$$

is holomorphic on D_M. Its zeros have an accumulation point there, so the identity theorem gives

$$\Theta'(z) = F(z)\Theta(z) \qquad (z \in D_M). \tag{6.2}$$

This passage does not assume that Θ is zero-free on D_M. The expression (6.1) has no division by Θ.

Suppose, for a contradiction, that Θ has a zero $z_0 \in D_M$. Since $\Theta(0) > 0$, it is not identically zero. Its zero at z_0 has some finite order $m \geq 1$, so near z_0,

$$\Theta(z) = (z - z_0)^m h(z), \qquad h(z_0) \neq 0,$$

where h is holomorphic. Substituting in (6.2) and cancelling $(z - z_0)^{m-1}$ off z_0 gives

$$mh(z) + (z - z_0)h'(z) = (z - z_0)F(z)h(z).$$

Both sides are continuous at z_0; evaluation there yields $mh(z_0) = 0$, a contradiction. Therefore every zero of Θ lies outside D_M, hence in $(-\infty, -1/M] \subset (-\infty, 0)$. Proposition 2.1 gives RH. Finally, restricting (6.2) to $a \geq 0$ and dividing by the already known positive $\Theta(a)$ proves $F(a) = L(a)$ everywhere on that ray. □

The proof uses neither a preselected logarithm of Θ on the cut plane nor an RH-dependent product. Choosing a global logarithm before proving zero-freeness would conceal the central difficulty. The division-free ODE avoids that issue.

6.2 The integration constant is real and must be treated

Suppose only the differential identity (1.5) holds. From (3.8),

$$\frac{\mathrm{d}}{\mathrm{d}a}(aL(a) - af(a)) = 0 \quad (a \in (\alpha, \beta)).$$

The ordinary mean value theorem on subintervals shows that

$$a(L(a) - F(a)) = C, \qquad L(a) = F(a) + \frac{C}{a} \tag{6.3}$$

for a constant $C \in \mathbb{R}$. Neither the derivative identity nor the off-diagonal Gram identity alone detects this constant: adding C to $aL(a)$ changes neither its derivative nor its divided differences.

For an arbitrary function on an interval bounded away from zero, one would need a normalization value to set $C = 0$. The actual entire Θ supplies a stronger analytic normalization.

Theorem 6.2 (Differential seed and normalized seed agree)**.** *For the actual Θ, $\mathsf{DSeed}_\Theta \iff \mathsf{HSeed}_\Theta$. The same operator and vector can be used in both directions.*

Proof. If HSeed_Θ holds, differentiating $aL(a) = af(a)$ and using (3.8) gives DSeed_Θ. Conversely assume DSeed_Θ and let C be the constant in (6.3). Consider

$$Q(z) = z\Theta'(z) - (zF(z) + C)\Theta(z). \tag{6.4}$$

This is holomorphic on all of D_M, including zero. By (6.3), it vanishes on (α, β), so the identity theorem gives $Q = 0$ throughout D_M. At zero,

$$0 = Q(0) = -C\Theta(0).$$

Equation (2.5) therefore forces $C = 0$. It follows that $z(\Theta' - F\Theta) = 0$ on D_M. Away from zero we may divide by z, and continuity supplies the equality at zero. Thus (6.2) holds, and positive-ray division by $\Theta(a)$ gives the normalized calibration. In particular HSeed_Θ holds. □

Corollary 6.3 (The recovered norm-square identity is an RH criterion)**.** *For a fixed bounded nonnegative self-adjoint K and fixed v, the identity (1.5) on any nonempty positive interval implies RH. No extra origin-normalization premise is needed beyond the established facts about the actual Θ.*

The qualifier "actual" is essential. If one replaced Θ by $z^m\Theta$ with $m > 0$, its logarithmic derivative would acquire m/z, and the derivative of aL would not see that origin factor. The known value $\Theta(0) \neq 0$ is precisely what rules out this ambiguity here.

7 The converse: constructing a seed under RH

7.1 Why the product has no exponential ambiguity

We use two classical facts about entire functions. The completed ξ has order at most 1; a direct growth argument from the theta integral is given in [4]. Hadamard factorization and its zero-summability consequence are stated in [5, Theorems 3.7 and 3.11]. To see exactly how these apply, write $M_E(r) = \max_{|w|=r} |E(w)|$ and define M_Θ similarly. The map $w \mapsto w^2$ maps the circle of radius $\sqrt{r}$ onto the circle of radius r, so

$$M_\Theta(r) = M_E(\sqrt{r}). \tag{7.1}$$

A bound $|E(w)| \le C_\varepsilon \exp(C_\varepsilon |w|^{1+\varepsilon})$ therefore puts the order of Θ at most $1/2$. Only the strict inequality "order less than 1" is needed below.

For an entire function of order less than 1, nonzero at zero, Hadamard factorization has genus zero and a constant exponential factor. Its nonzero zeros z_n, repeated by multiplicity, satisfy $\sum_n |z_n|^{-1} < \infty$. Hence

$$\Theta(z) = \Theta(0) \prod_n \left(1 - \frac{z}{z_n}\right). \tag{7.2}$$

There is no factor e^{bz} with $b \neq 0$: the polynomial in the Hadamard exponential has degree at most the order, hence degree zero. There is also no origin monomial because $\Theta(0) \neq 0$.

7.2 Zero pairs and multiplicity

Assume RH. Enumerate the positive ordinates of the nontrivial zeros as $\gamma_1, \gamma_2, \ldots$, repeating each according to its multiplicity. There are infinitely many zeros, with no finite accumulation point [3]. We can arrange $0 < \gamma_1 \le \gamma_2 \le \cdots$ with $\gamma_n \to \infty$. The two points $w = \pm i\gamma$ are the preimages of $z = -\gamma^2$. Since the derivative of $w \mapsto w^2$ is nonzero at either preimage, a zero of multiplicity m at either point corresponds to a zero of multiplicity m of Θ. It does *not* become a zero of multiplicity $2m$. Equation (7.2) becomes

$$\Theta(z) = \Theta(0) \prod_{n=1}^{\infty} \left(1 + \frac{z}{\gamma_n^2}\right), \qquad \sum_{n=1}^{\infty} \frac{1}{\gamma_n^2} < \infty. \tag{7.3}$$

Thus one positive/negative pair contributes one factor for each occurrence of its multiplicity.

7.3 Logarithmic differentiation with convergence controlled

Fix a compact set disjoint from the negative zero set, and enclose it in $|z| \le r$. For all sufficiently large n, $\gamma_n^2 > 2r$ and

$$\left|\frac{1}{z + \gamma_n^2}\right| \le \frac{2}{\gamma_n^2}.$$

The right side is summable by (7.3). Finite initial factors have no zeros on the chosen compact set. Locally uniform convergence of the product and its logarithmic derivatives therefore justifies

$$\frac{\Theta'(z)}{\Theta(z)} = \sum_{n=1}^{\infty} \frac{1}{z + \gamma_n^2} \tag{7.4}$$

away from the zeros. In particular this identity holds at every $a \ge 0$.

Proposition 7.1 (Diagonal realization under RH)**.** *Let $\mathcal{H} = \ell^2(\mathbb{N}; \mathbb{C})$, and define*

$$(Kx)_n = \gamma_n^{-2} x_n, \qquad v_n = \gamma_n^{-1}. \tag{7.5}$$

Then K is bounded, self-adjoint, nonnegative, compact and trace class; $v \in \mathcal{H}$; and

$$\langle v, (I + aK)^{-1} v \rangle = L(a) \qquad (a \geq 0). \tag{7.6}$$

Proof. The positive ordinates have a positive minimum: a sequence tending to zero would give an accumulation of zeros at $1/2$, contradicting $\xi(1/2) > 0$. Hence the diagonal entries of K are bounded. They are positive real numbers and tend to zero, which gives self-adjointness, positivity and compactness. Their sum is finite by (7.3), so the positive diagonal operator is trace class. The same sum equals $\sum_n |v_n|^2$, proving $v \in \mathcal{H}$. Finally,

$$\langle v, (I + aK)^{-1} v \rangle = \sum_n \frac{\gamma_n^{-2}}{1 + a\gamma_n^{-2}} = \sum_n \frac{1}{a + \gamma_n^2} = L(a),$$

using (7.4). □

This proves RH $\Rightarrow \mathsf{HSeed}_\Theta$, completing Theorem 1.3. Taking $a = 0$ gives the normalization

$$\operatorname{Tr} K = \|v\|^2 = \sum_n \gamma_n^{-2} = L(0) = \frac{\xi''(1/2)}{2\xi(1/2)}. \tag{7.7}$$

For an arbitrary calibrated representation, $\|v\|^2 = L(0)$ still holds. The equality with $\operatorname{Tr} K$ is asserted only for the particular diagonal construction just given; arbitrary invisible operator components need not have that trace.

If one prefers distinct positive ordinates γ_j with multiplicities m_j, the equivalent scalar model uses $K_{jj} = \gamma_j^{-2}$ and $v_j = \sqrt{m_j}/\gamma_j$. Neither version assumes that zeta zeros are simple. In the repeated model, the vector need not be cyclic when an eigenvalue is repeated. No cyclicity assumption is needed for the argument.

8 Spectral measures, moments and exact reconstruction

8.1 A scalar measure formulation

Define MSeed_Θ to mean that for some $M > 0$ and finite positive Borel measure μ on $[0, M]$,

$$L(a) = \int_{[0,M]} \frac{1}{1 + at} \, \mathrm{d}\mu(t) \tag{8.1}$$

on a nonempty positive interval. Section 5 proves $\mathsf{HSeed}_\Theta \Rightarrow \mathsf{MSeed}_\Theta$. Conversely, given (8.1), take

$$\mathcal{H} = L^2([0, M], \mu), \qquad (Ku)(t) = tu(t), \qquad v(t) = 1.$$

Finite mass makes v square integrable, while multiplication by t is bounded, nonnegative and self-adjoint. Its scalar resolvent is exactly (8.1). Therefore

$$\mathsf{MSeed}_\Theta \iff \mathsf{HSeed}_\Theta \iff \text{RH}. \tag{8.2}$$

The word "finite" in *finite measure* means finite total mass. It does not mean finite support, a finite list of rational numbers, or a finite-checker witness. Under RH the measure is

$$\mu = \sum_n \gamma_n^{-2} \delta_{\gamma_n^{-2}}, \qquad \mu([0, M]) = \kappa. \tag{8.3}$$

This differs from the zero-counting measure $\sum_n \delta_{\gamma_n^2}$, which has infinite total mass. The reciprocal variable and the weights are essential to the normalization.

8.2 Moments and positive finite sections

For any fixed positive operator model, put

$$m_n = \langle v, K^n v\rangle = \int_{[0,M]} t^n \, \mathrm{d}\mu(t) \qquad (n \geq 0). \tag{8.4}$$

If $|z|M < 1$, the Neumann series converges uniformly in the spectral variable and gives

$$F(z) = \sum_{n=0}^{\infty} (-1)^n m_n z^n, \qquad 0 \leq m_n \leq M^n m_0. \tag{8.5}$$

The remainder after degree N has the exact form

$$F(z) - \sum_{n=0}^{N} (-1)^n m_n z^n = (-z)^{N+1} \int \frac{t^{N+1}}{1+zt} \, \mathrm{d}\mu(t), \tag{8.6}$$

and consequently

$$\left| F(z) - \sum_{n=0}^{N} (-1)^n m_n z^n \right| \leq \frac{m_0 (M|z|)^{N+1}}{1 - M|z|}. \tag{8.7}$$

For a calibrated model,

$$m_n = \frac{(-1)^n}{n!} L^{(n)}(0). \tag{8.8}$$

This is an all-orders identity for actual derivatives.

For a polynomial $p(t) = \sum_i c_i t^i$, positivity of μ gives

$$\sum_{i,j} \overline{c_i} c_j m_{i+j} \geq 0, \tag{8.9}$$

$$\sum_{i,j} \overline{c_i} c_j m_{i+j+1} \geq 0, \tag{8.10}$$

$$\sum_{i,j} \overline{c_i} c_j (M^2 m_{i+j} - m_{i+j+2}) \geq 0. \tag{8.11}$$

These are the integrals of $|p|^2$, $t|p|^2$ and $(M^2 - t^2)|p|^2$. Every finite matrix in this hierarchy is positive semidefinite. The following converse records exactly what an *all-orders* moment construction can accomplish.

Theorem 8.1 (Bounded positive moment reconstruction)**.** *Let $(m_n)_{n\geq 0}$ be a real sequence and $M > 0$. Suppose (8.9)–(8.11) hold for every finite coefficient list. Then there exist a Hilbert space, a bounded nonnegative self-adjoint operator K with $\|K\| \leq M$, and a vector v such that $m_n = \langle v, K^n v\rangle$ for all n.*

Proof. Define a complex linear functional ℓ on polynomials by $\ell(t^n) = m_n$, and a sesquilinear form by $\langle p, q\rangle_0 = \ell(\overline{p}\, q)$, where conjugation acts on coefficients and t is real. Condition (8.9) makes this form positive semidefinite, hence it satisfies Cauchy–Schwarz. Quotient the polynomial space by the vectors of zero seminorm and complete to a Hilbert space $\mathcal{H}$.

Multiplication by t is well defined on the quotient: by (8.11),

$$\|tp\|_0^2 = \ell(t^2|p|^2) \leq M^2 \ell(|p|^2) = M^2 \|p\|_0^2.$$

Thus it sends every null vector to a null vector and is bounded by M. It extends to an everywhere defined bounded operator K on $\mathcal{H}$. Since $\ell(\overline{tp}\, q) = \ell(\overline{p}\, tq)$, it is symmetric on a dense subspace and hence self-adjoint after extension. Condition (8.10) gives $\langle [p], K[p]\rangle = \ell(t|p|^2) \geq 0$; continuity makes K positive on all of $\mathcal{H}$. Let $v = [1]$. Then $K^n v = [t^n]$ and $\langle v, K^n v\rangle = \ell(t^n) = m_n$, as required. □

Corollary 8.2 (An all-order actual-jet criterion)**.** *Define m_n by (8.8). RH is equivalent to the existence of $M > 0$ such that (8.9)–(8.11) hold for every finite coefficient list.*

Proof. Under RH, the diagonal seed supplies these inequalities. Conversely, Theorem 8.1 constructs K, v with the specified moments. The resolvent has Taylor series (8.5). The actual L is analytic near zero and has exactly the same Taylor coefficients. They therefore agree on a common disk of positive radius, and in particular on a positive open interval. This is HSeed_Θ, which implies RH. □

The quantifier "for every finite coefficient list" ranges over unbounded degrees. Checking finitely many matrices does not establish this corollary. A finite certificate would have to derive the entire hierarchy by an independently proved uniform mechanism.

8.3 What is determined by calibration

If two compactly supported finite positive measures represent the same actual L on an interval, analytic continuation gives equality near zero. Their moments consequently agree. View both on a common compact interval containing their supports. Polynomials are dense in the continuous functions on that interval, and equality of integrals for polynomials extends, by the total-mass bound, to all continuous functions. Uniqueness of finite Borel measures then gives equality of the two measures. Thus the scalar spectral measure seen by v is determined. This does not identify every operator on the ambient Hilbert space: an orthogonal summand invisible to v leaves all the scalar identities unchanged.

9 Basepoint shifts and finite truncations

9.1 Two different shifted vectors

Fix $a_0 \geq 0$, and abbreviate $R_0 = (I + a_0K)^{-1}$ in this subsection only. Set

$$B = KR_0, \qquad w = R_0^{1/2}v, \qquad u = R_0v. \tag{9.1}$$

These two vectors have different purposes. For $t = a - a_0$ with $a \geq 0$,

$$I + aK = (I + a_0K)(I + tB).$$

All the displayed operators commute. Therefore

$$f(a_0 + t) = \langle w, (I + tB)^{-1}w\rangle. \tag{9.2}$$

The functional calculus gives $0 \leq B \leq M(1 + a_0M)^{-1}I$ whenever $0 \leq K \leq MI$. In particular the local shifted inverse is analytic at $t = 0$.

On the other hand, (3.4) gives

$$g(a_0 + t) - g(a_0) = t\langle u, (I + tB)^{-1}u\rangle. \tag{9.3}$$

To verify this, write $aR_a - a_0R_0 = tR_aR_0$ and $R_a = (I + tB)^{-1}R_0$, then take the scalar product with v. Differentiating (9.3) gives

$$g'(a_0 + t) = \|(I + tB)^{-1}u\|^2. \tag{9.4}$$

The additive value $g(a_0)$ and the distinction $w \neq u$ in general must be retained. Merely replacing a by $a - a_0$ in part of a formula changes the mathematical condition.

9.2 What finite diagonal truncations do prove

Under RH, use the repeated-ordinate model from Proposition 7.1, and truncate after N entries:

$$F_N(z) = \sum_{n=1}^{N} \frac{1}{z+\gamma_n^2}, \qquad S_N = \sum_{n>N} \gamma_n^{-2}.$$

For every $a \geq 0$,

$$0 \leq L(a) - F_N(a) \leq S_N. \tag{9.5}$$

To state a bound on a whole disk, define the tail directly by

$$T_N(z) = \sum_{n>N} \frac{1}{z+\gamma_n^2}.$$

If $\gamma_{N+1}^2 > 2r$, this series is holomorphic on a neighborhood of $|z| \leq r$. Away from the initial poles it equals $F - F_N$; at a shared pole among the first N terms, it is the holomorphic extension of that difference after cancellation. Thus, on the entire disk,

$$|T_N(z)| \leq 2S_N. \tag{9.6}$$

For each derivative order $j \geq 0$, the same argument gives

$$|T_N^{(j)}(z)| \leq j!\, 2^{j+1} \sum_{n>N} \gamma_n^{-2j-2}. \tag{9.7}$$

These estimates follow term by term from the convergent series and its displayed denominator bounds. They show what a tail certificate would need to control, including its dependence on derivative order.

The construction here assumes RH to identify the positive spectrum. It is consequently not an unconditional finite computation proving RH. Even a uniform small error does not imply exact equality; a uniform error bound for values alone does not automatically supply an error bound for all derivatives on the same domain.

10 Hyperbolic zeta identities and the actual-seed transfer

The hyperbolic zeta identities provide explicit positive measures from which the operator calculus of Section 3 can be constructed. This section makes that connection precise, including the normalization of the odd-zeta formulas. It also identifies a limitation: the resulting model functions have a different large-parameter behavior from the actual L. Their positivity is useful structural information, but these particular models cannot themselves be its exact calibration. We then derive an actual-ξ Mellin identity and a conditional coefficient-limit theorem for positive determinant models obtained from the project's signed pencils. The limit condition is stated as an open analytic obligation.

The source identities below are traced to the manuscript [11] and repository revision `b56c810` [12, 13, 14, 16, 17]. Only the displayed identities are used, with proofs or explicit classical inputs supplied here. Source inspection is separate from native compilation and axiom auditing.

10.1 A positive weight and its Mellin moments

For an integer $k \geq 0$, set

$$B(x) = \frac{1}{e^x - 1}, \qquad \omega_k(y) = \frac{2(2\pi)^{2k}}{(2k)!} y^{2k+1} B^{(2k)}(2\pi y), \qquad y > 0. \tag{10.1}$$

Here $B^{(2k)}$ denotes differentiation with respect to the argument of B, not with respect to y. This distinction determines the factor $(2\pi)^{2k}$. The weights and their moment identities are the relevant positive-measure objects in the hyperbolic zeta development [12, 13]. If a continuous weight parameter $s > 1$ is used, the coefficient extending these weights is $2(2\pi)^{s-1}/\Gamma(s)$. The printed continuous coefficient $(2\pi)^s/(2\Gamma(s))$ differs by $\pi/2$; the repository records this correction explicitly [15].

Proposition 10.1 (Positivity and exact moments)**.** *The function ω_k is strictly positive, tends to $1/\pi$ at zero, and defines a finite measure on $(0,\infty)$. For every real $q > -1$,*

$$\int_0^\infty y^q \omega_k(y)\,\mathrm{d}y = \frac{2\Gamma(q+2k+2)\zeta(q+2)}{(2k)!(2\pi)^{q+2}}. \tag{10.2}$$

In particular,

$$\int_0^\infty \omega_k(y)\,\mathrm{d}y = \frac{2k+1}{12}. \tag{10.3}$$

Proof. The geometric series and its locally uniformly convergent derivatives give

$$B^{(2k)}(x) = \sum_{n=1}^{\infty} n^{2k} e^{-nx} > 0 \qquad (x > 0).$$

The Laurent expansion $B(x) = x^{-1} - 1/2 + O(x)$, with its analytic remainder near zero, shows $B^{(2k)}(x) \sim (2k)!x^{-2k-1}$. Substitution in (10.1) gives $\omega_k(y) \to 1/\pi$. At infinity the differentiated geometric series is $O(e^{-x})$, so the weight has exponential decay times a polynomial.

For the exact value, Tonelli's theorem applies because every summand is nonnegative. The gamma integral then yields

$$\begin{aligned}\int_0^\infty y^q \omega_k(y)\,\mathrm{d}y &= \frac{2(2\pi)^{2k}}{(2k)!}\sum_{n\ge1} n^{2k}\frac{\Gamma(q+2k+2)}{(2\pi n)^{q+2k+2}} \\ &= \frac{2\Gamma(q+2k+2)}{(2k)!(2\pi)^{q+2}}\sum_{n\ge1} n^{-q-2}.\end{aligned}$$

The last series converges precisely in the stated real range. Taking $q = 0$ and using $\zeta(2) = \pi^2/6$ proves (10.3). □

Thus polynomial Gram matrices have exact entries involving even and odd zeta values. Their positivity alone does not make multiplication by y or y^2 bounded: the measure has unbounded support. The next construction supplies the bounded operator required by the seed framework. For instance, the even moments $b_j = \int y^{2j}\omega_k(y)\,\mathrm{d}y$ satisfy

$$\frac{b_{j+1}}{b_j} = \frac{(2k+2j+3)(2k+2j+2)}{4\pi^2}\frac{\zeta(2j+4)}{\zeta(2j+2)} \longrightarrow \infty. \tag{10.4}$$

Thus no fixed finite support bound can be read into the raw moment sequence. Positivity and bounded support are different requirements in the reconstruction theorem. Here the support is made compact by an explicit change of spectral variable, whose effect on the vector and measure must also be included.

10.2 Compactification and the fixed Hilbert vector

Put

$$t(y) = \frac{1}{1+y^2}, \qquad \mathcal{H}_k = L^2((0,\infty), \omega_k(y)\,\mathrm{d}y), \qquad (K_k u)(y) = t(y)u(y), \qquad v_k(y) = \sqrt{t(y)}. \tag{10.5}$$

Then K_k is nonnegative and self-adjoint, with $\|K_k\| = 1$. Indeed, $0 < t(y) < 1$, and values arbitrarily close to 1 occur on sets of positive measure. Also $v_k \in \mathcal{H}_k$ by (10.3). Its scalar resolvent is

$$F_k(a) = \langle v_k, (I + aK_k)^{-1} v_k \rangle = \int_0^\infty \frac{\omega_k(y)}{a + 1 + y^2}\, \mathrm{d}y, \qquad a > -1. \tag{10.6}$$

In particular, $\|v_k\|^2 = F_k(0)$; it is not the total mass in (10.3). The factor $\sqrt{t(y)}$ in the vector is essential to this normalization: its square supplies the numerator $t(y)$ in $t(y)/(1 + at(y)) = (a+1+y^2)^{-1}$. All parameters consequently use one operator and one vector, exactly as required in the seed definitions. The construction is infinite dimensional, despite the single integer parameter specifying the density.

Define the finite measure ν_k on $[0, 1]$ by

$$\int \varphi(t)\, \mathrm{d}\nu_k(t) = \int_0^\infty t(y)\varphi(t(y))\omega_k(y)\, \mathrm{d}y.$$

Its moments are

$$q_n^{(k)} = \int_0^1 t^n\, \mathrm{d}\nu_k(t) = \int_0^\infty \frac{\omega_k(y)}{(1 + y^2)^{n+1}}\, \mathrm{d}y, \qquad F_k(a) = \sum_{n \geq 0} (-1)^n q_n^{(k)} a^n \quad (|a| < 1). \tag{10.7}$$

The series follows from a uniformly convergent geometric series on $0 \leq t \leq 1$. For every polynomial $p(t) = \sum_i c_i t^i$, integration of $|p|^2$, $t|p|^2$, $(1 - t)|p|^2$ and $(1 - t^2)|p|^2$ proves positivity of all finite sections of

$$(q_{i+j}^{(k)}), \quad (q_{i+j+1}^{(k)}), \quad (q_{i+j}^{(k)} - q_{i+j+1}^{(k)}), \quad (q_{i+j}^{(k)} - q_{i+j+2}^{(k)}). \tag{10.8}$$

These are all-order identities for this explicitly defined model. They make no identification with the actual moments $(-1)^n L^{(n)}(0)/n!$. There is also the full compact-moment difference identity

$$\sum_{j=0}^{r} (-1)^j \binom{r}{j} q_{n+j}^{(k)} = \int_0^1 t^n (1 - t)^r\, \mathrm{d}\nu_k(t) \geq 0 \qquad (n, r \geq 0). \tag{10.9}$$

For a nonzero polynomial, each quadratic form in (10.8) is strictly positive: its integrand is positive except at finitely many roots and possibly the endpoints, and the transformed measure gives positive mass to every open subinterval of $(0, 1)$.

The compact moments should not be obtained by formally expanding $(1+y^2)^{-1}$ into powers of y^2 and integrating term by term. That geometric series does not converge throughout the original unbounded support. Formula (10.7), or the Hurwitz calculation below, supplies a justified route instead.

With $R_{k,a} = (I + aK_k)^{-1}$ and $g_k(a) = aF_k(a)$, the general resolvent calculation gives the explicit formulas

$$g_k'(a) = \|R_{k,a} v_k\|^2 = \int_0^\infty \frac{(1 + y^2)\omega_k(y)}{(a + 1 + y^2)^2}\, \mathrm{d}y, \tag{10.10}$$

$$\frac{g_k(a) - g_k(b)}{a - b} = \langle R_{k,a} v_k, R_{k,b} v_k \rangle = \int_0^\infty \frac{(1 + y^2)\omega_k(y)}{(a + 1 + y^2)(b + 1 + y^2)}\, \mathrm{d}y. \tag{10.11}$$

The second formula is for distinct nonnegative a, b and extends diagonally by the first. Multiplication by $4\sqrt{ab}$ produces the Gram identity for $2\sqrt{a}\, R_{k,a} v_k$.

10.3 The Hurwitz identity and its endpoint normalization

Proposition 10.2 (Real-parameter odd-zeta resolvent)**.** *For $k \geq 1$, $a > -1$, and $z = \sqrt{1 + a} > 0$,*

$$F_k(a) = z^{2k} \zeta(2k + 1, z) - \frac{1}{2z} - \frac{1}{2k}. \tag{10.12}$$

For $k = 0$,

$$F_0(a) = \log z - \frac{1}{2z} - \psi(z). \tag{10.13}$$

Proof. Binet's digamma formula [18] states that, for real $z > 0$,

$$h(z) := \log z - \frac{1}{2z} - \psi(z) = 2\int_0^\infty \frac{yB(2\pi y)}{y^2+z^2}\,\mathrm{d}y. \tag{10.14}$$

This immediately proves the case $k = 0$. For $k \geq 1$, use $\partial_y^{2k} B(2\pi y) = (2\pi)^{2k} B^{(2k)}(2\pi y)$ and integrate by parts $2k$ times:

$$F_k(z^2-1) = \frac{2}{(2k)!}\int_0^\infty B(2\pi y)\partial_y^{2k}\left(\frac{y^{2k+1}}{y^2+z^2}\right)\mathrm{d}y.$$

Every boundary term vanishes. At zero, the product of the relevant derivatives is $O(y)$: the rational factor has a zero of order $2k+1$, whereas $\partial_y^j B(2\pi y) = O(y^{-j-1})$. At infinity the exponential decay dominates the polynomial factors.

Polynomial division gives

$$\frac{y^{2k+1}}{y^2+z^2} = P_{2k-1}(y) + (-z^2)^k \frac{y}{y^2+z^2}.$$

The polynomial disappears after $2k$ differentiations. From $y/(y^2+z^2) = \frac{1}{2}((y+iz)^{-1}+(y-iz)^{-1})$ one obtains

$$\partial_y^{2k}\frac{y}{y^2+z^2} = (-1)^k\partial_z^{2k}\frac{y}{y^2+z^2}.$$

On compact positive z-intervals these integral derivatives are dominated near zero and at infinity. Consequently

$$F_k(z^2-1) = \frac{z^{2k}}{(2k)!}h^{(2k)}(z). \tag{10.15}$$

More explicitly, every fixed z-derivative of $y/(y^2+z^2)$ is $O(y)$ near $y = 0$, uniformly when z stays away from zero. Multiplication by $B(2\pi y) = O(y^{-1})$ therefore gives a locally integrable uniform bound. At infinity, the Bose factor supplies exponential domination. This verifies the exchange of differentiation and integration used in (10.15). Finally, $\psi^{(2k)}(z) = -(2k)!\zeta(2k+1,z)$ [19]; differentiating the logarithm and reciprocal in h supplies respectively $-1/(2k)$ and $-1/(2z)$ after multiplication by $z^{2k}/(2k)!$. This proves (10.12). □

Taking values and first derivatives at $a = 0$ gives

$$\begin{aligned} q_0^{(k)} &= \zeta(2k+1) - \frac{1}{2} - \frac{1}{2k}, \\ q_1^{(k)} &= \frac{2k+1}{2}\zeta(2k+2) - k\zeta(2k+1) - \frac{1}{4} \qquad (k \geq 1). \end{aligned} \tag{10.16}$$

For $k = 0$ the corresponding values are $q_0^{(0)} = \gamma - 1/2$ and $q_1^{(0)} = \zeta(2)/2 - 3/4$, where γ is Euler's constant. The differentiation uses $\partial_a = (2z)^{-1}\partial_z$ and $\partial_z\zeta(s,z) = -s\zeta(s+1,z)$. For $k \geq 1$, repetition expresses every fixed compact moment through finitely many adjacent zeta values and rational constants. The integral formulas guarantee, independently of any cancellation in those expressions, that $q_0^{(k)} > q_1^{(k)} > q_2^{(k)} > \cdots > 0$. This gives exact positivity identities suitable for comparing proposed moment constructions, without supplying the actual Θ jets.

The integer-parameter versions must retain their endpoint convention [14]. For $k \geq 1$, $N \geq 1$, and $s = 2k + 1$,

$$\begin{aligned} F_k(N^2 - 1) &= N^{2k} \left(\zeta(s) - \sum_{n=1}^{N-1} n^{-s} \right) - \frac{1}{2N} - \frac{1}{2k} \\ &= N^{2k} \left(\zeta(s) - \sum_{n=1}^{N} n^{-s} \right) + \frac{1}{2N} - \frac{1}{2k}. \end{aligned} \tag{10.17}$$

Including the Nth summand changes the sign of the displayed $1/(2N)$ contribution. At $N = 1$, either version recovers the value of $q_0^{(k)}$ in (10.16).

10.4 An exact Bose term in the actual logarithmic derivative

There is a direct identity involving the actual completed zeta function. Let $a > 1/4$ and $s = 1/2 + \sqrt{a}$, so $s > 1$. Logarithmic differentiation of (1.2), Binet's formula, and the convergent von Mangoldt series [18, 20] give

$$\begin{aligned} 2\sqrt{a}\, L(a) = \frac{1}{2s} + \frac{1}{s-1} + \frac{1}{2} \log \frac{s}{2\pi} \\ - \int_0^\infty \frac{y \, \mathrm{d}y}{(y^2 + s^2/4)(e^{2\pi y} - 1)} - \sum_{n=2}^{\infty} \Lambda(n) n^{-s}. \end{aligned} \tag{10.18}$$

Indeed, before substitution the gamma contribution is $\frac{1}{2}\psi(s/2)$ and the elementary contribution includes $1/s$. The complete starting identity is

$$\frac{\xi'(s)}{\xi(s)} = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2}\psi(s/2) + \frac{\zeta'(s)}{\zeta(s)}.$$

Binet contributes $-1/(2s)$, leaving the coefficient $1/(2s)$ in (10.18). The same Bose kernel therefore occurs in the actual function, but together with the logarithmic, rational, and prime contributions. The positivity of one integral does not establish a positive-resolvent representation of their complete sum.

Proposition 10.3 (Asymptotic obstruction to direct calibration)**.** *No F_k equals the actual L on a nonempty open positive interval. More generally, this remains impossible for a finite sum $G(a) = \sum_{j=1}^{J} c_j F_{k_j}(d_j a + e_j)$ with $c_j, d_j > 0$ and $e_j \geq 0$.*

Proof. Dominated convergence in (10.6) gives

$$\lim_{a \to \infty} a F_k(a) = \frac{2k+1}{12}, \qquad \lim_{a \to \infty} a G(a) = \sum_{j=1}^{J} \frac{c_j (2k_j + 1)}{12 d_j} < \infty. \tag{10.19}$$

For the actual function, the integral in (10.18) is bounded by $1/(6s^2)$, using $\int_0^\infty y B(2\pi y)\, \mathrm{d}y = 1/24$. The prime sum is exponentially small as $s \to \infty$: for $s \geq 4$ it is bounded by $2^{-s/2} \sum_{n \geq 2} (\log n) n^{-2}$. The remaining rational terms are $O(s^{-1})$. Since $s = 1/2 + \sqrt{a}$, it follows that

$$L(a) = \frac{\log a}{8\sqrt{a}} - \frac{\log(2\pi)}{4\sqrt{a}} + O(a^{-1}), \qquad a L(a) \longrightarrow +\infty. \tag{10.20}$$

Both G and L are real analytic throughout $(0, \infty)$. Equality on an open subinterval would imply equality on that connected ray, contradicting (10.19) and (10.20). □

This obstruction is consistent with the abstract Hilbert-seed theorem. A finite compact spectral measure need not have a finite reciprocal moment. For the present models, however,

$$\int_{(0,1]} \frac{1}{t}\,\mathrm{d}\nu_k(t) = \int_0^\infty \omega_k(y)\,\mathrm{d}y = \frac{2k+1}{12}. \tag{10.21}$$

This extra integrability is exactly what makes $aF_k(a)$ tend to a finite limit. Under RH, the canonical measure in (8.3) instead has reciprocal moment $\sum_n 1 = \infty$, although its total mass is finite. Thus one cannot reject all bounded positive operators from Proposition 10.3. The result rejects this particular specific Bose multiplication construction and its stated finite combinations. It identifies a concrete property that a different construction would have to change.

Similarly, the positive matrices in (10.8) can serve as exact examples for checking normalization, shifts and proposed reconstruction lemmas. Their proofs work uniformly in degree because a specific positive measure was supplied first. Transferring that conclusion to the actual jets would require a theorem equating the entire moment sequences, not agreement of several low-order entries or a numerical fit on an interval.

The usable conclusion is precise: the hyperbolic identities supply explicit positive models, exact moment hierarchies, and a Bose integral inside the actual calibration problem. The bounded compactification above does not solve that problem. Any further construction must account for the full expression (10.18), including the growth in (10.20), before it can satisfy HSeed_Θ.

10.5 From hyperbolic source identities to the actual logarithmic moments

The useful connection begins with an identity for the actual completed function, rather than with a positive model chosen independently of it. Write $B(x) = (e^x - 1)^{-1}$ for $x > 0$. Elementary cancellation gives the positive density

$$h(x) := 4B(2x) - 8B(4x) = \frac{4}{e^{2x}+1} = 2(1 - \tanh x) > 0. \tag{10.22}$$

Thus a difference of two Bose kernels is exactly a positive Fermi, or hyperbolic-tangent, kernel. Its positivity follows after the cancellation; the two separate terms should not be treated as a positive-measure decomposition.

Define the Mellin integral

$$I(s) = \int_0^\infty x^{s-1} h(x)\,\mathrm{d}x = 2^{2-s}\Gamma(s)(1 - 2^{1-s})\zeta(s), \qquad \Re s > 0. \tag{10.23}$$

Here the expression at $s = 1$ is removable, with $I(1) = 2\log 2$. For $\Re s > 1$, expand $(e^{2x}+1)^{-1} = \sum_{k\ge1}(-1)^{k-1}e^{-2kx}$. Absolute convergence after integration permits termwise use of the gamma integral and gives $I(s) = 2^{2-s}\Gamma(s)\sum_{k\ge1}(-1)^{k-1}k^{-s}$. Separating even and odd terms proves (10.23) there. Both sides continue holomorphically to $\Re s > 0$: on the integral side, $h(x) = O(1)$ at zero and $h(x) = O(e^{-2x})$ at infinity; on the function side, the factor $1 - 2^{1-s}$ removes the pole of ζ at 1. The identity theorem gives the claimed domain. This is the scaled Fermi integral in [21, Eq. 25.5.3].

The same density is obtained from the explicitly defined two-variable measure

$$q(r) = \frac{r}{e^r - 1}, \qquad d(r) = -q'(r) > 0, \qquad \mathrm{d}\mu(r,y) = 2d(r)\,\mathrm{d}r\,\frac{\mathrm{d}y}{y}, \quad \frac{1}{4} \le y \le \frac{1}{2}, \tag{10.24}$$

by pushing forward under $x = ry$. Indeed its density is

$$2\int_{1/4}^{1/2} d(x/y)\,\frac{\mathrm{d}y}{y^2} = \frac{2}{x}(q(2x) - q(4x)) = h(x).$$

For completeness, $d(r) > 0$ because $-q'(r) = [e^r(r-1)+1]/(e^r-1)^2$, and the numerator vanishes at zero and has derivative $re^r > 0$. This calculation gives an exact one-variable realization of the source layer. In particular its total mass is $I(1) = 2\log 2$, and the original two-variable integral of $(ry)^{s-1}$ is precisely $I(s)$. Tonelli's theorem applies first to nonnegative integrands; absolute integrability then gives the same change of variables for the complex Mellin integrands used here.

The gamma duplication formula [22] simplifies the normalizing factor to

$$A(s) = \frac{s(s-1)\pi^{(1-s)/2}}{4\Gamma((s+1)/2)(1-2^{1-s})}, \qquad \xi(s) = A(s)I(s). \tag{10.25}$$

The equality holds in a complex neighborhood of $s = 1/2$, which is the domain needed here. The denominator is nonzero there, and A is positive on a real neighborhood of $1/2$: both $s(s-1)$ and $1-2^{1-s}$ are negative for $0 < s < 1$. We do not claim that the displayed A is holomorphic throughout the right half-plane; its other apparent singularities require cancellation in the product.

10.5.1 An explicit map from the source integrals to the required moments

Differentiation under the integral is valid to every fixed order:

$$I_j(s) := I^{(j)}(s) = \int_0^\infty x^{s-1}(\log x)^j h(x)\,\mathrm{d}x. \tag{10.26}$$

On each compact substrip $\Re s \geq \sigma > 0$, the behavior at zero is controlled by $x^{\sigma-1}|\log x|^j$, and the exponential tail controls infinity. In particular all the integrals at $1/2$ are absolutely convergent. The product rule now gives actual, unconditional coefficient identities:

$$a_n := \frac{\xi^{(2n)}(1/2)}{(2n)!\,\xi(1/2)}, \qquad a_0 = 1, \tag{10.27}$$

$$\xi^{(2n)}(1/2) = \sum_{j=0}^{2n} \binom{2n}{j} A^{(2n-j)}(1/2) I_j(1/2), \tag{10.28}$$

$$\frac{\Theta(z)}{\Theta(0)} = \sum_{n\geq 0} a_n z^n. \tag{10.29}$$

Consequently, hyperbolic identities that evaluate or bound these integrals can be transported to the actual Θ coefficients with every normalization factor visible.

The resolvent moments are different quantities. Set $m_n = (-1)^n L^{(n)}(0)/n!$ as before. Comparing coefficients in $\Theta' = L\Theta$ gives the exact triangular recurrence

$$(n+1)a_{n+1} = \sum_{k=0}^{n} (-1)^k m_k a_{n-k} \qquad (n \geq 0). \tag{10.30}$$

It determines every required m_n from $a_1, \ldots, a_{n+1}$. The first cases and the first nontrivial Hankel determinant are

$$\begin{aligned} m_0 &= a_1, \\ m_1 &= a_1^2 - 2a_2, \\ m_2 &= a_1^3 - 3a_1a_2 + 3a_3, \end{aligned} \tag{10.31}$$

$$\det \begin{pmatrix} m_0 & m_1 \\ m_1 & m_2 \end{pmatrix} = a_1^2 a_2 + 3a_1 a_3 - 4a_2^2. \tag{10.32}$$

These are algebraic identities, not positivity assertions. In particular, positive formulas for the individual a_n do not settle the signs of their nonlinear combinations. To check the recurrence directly, multiply $\sum_{k\ge0}(-1)^k m_k z^k$ by $\sum_{j\ge0} a_j z^j$ in a sufficiently small disk where Θ is nonzero. The coefficient of z^n is the right side of (10.30), whereas differentiation of the second series gives its left side. The occurrence of m_n with coefficient $(-1)^n a_0 = (-1)^n$ makes the recursion unique.

There is also a compact cumulant interpretation. Let $X = \log x$ under the probability measure $\mathrm{d}\nu = x^{-1/2}h(x)\,\mathrm{d}x/I(1/2)$, and write $\mathrm{cum}_j(X)$ for its jth cumulant. Its moment generating function exists near zero, and $\mathrm{cum}_j(X) = (\log I)^{(j)}(1/2)$. Indeed $\mathbf{E}_\nu[e^{tX}] = I(1/2+t)/I(1/2)$ for $|\Re t| < 1/2$. The preceding endpoint estimates control this identity and all its fixed derivatives on smaller strips. Evenness of $\xi(1/2+w)$ and its nonzero value at zero give

$$m_n = \frac{(-1)^n}{2(2n+1)!}\Big((\log A)^{(2n+2)}(1/2) + \mathrm{cum}_{2n+2}(X)\Big). \tag{10.33}$$

All logarithms here are local analytic logarithms chosen real on the relevant real interval. Formula (10.33) shows explicitly why ordinary positive-measure moments do not automatically yield the required alternating logarithmic moments. The derivatives of $\log A$ are part of the calibration and cannot be discarded. For example, the second derivative of $\log I$ is the variance of X, but the second derivative of $\log\xi$ also contains $(\log A)''$. Higher cumulants have no positivity law comparable to the elementary squared-norm identity for ordinary moment matrices.

10.5.2 The positive mass and the remaining positivity theorem

The mass $\kappa = m_0$ is strictly positive without RH. One standard positive Riemann kernel, obtained by rescaling the representation in [23, Section 4, Eqs. (4.1)–(4.2)], is

$$\Psi(u) = 4\sum_{n\ge1}\pi n^2\big(2\pi n^2 e^{2u} - 3\big)e^{5u/2-\pi n^2 e^{2u}}, \qquad u \ge 0, \tag{10.34}$$

$$\xi(1/2+w) = \int_0^\infty \Psi(u)\cosh(wu)\,\mathrm{d}u. \tag{10.35}$$

Every summand is positive, since $2\pi n^2 e^{2u} > 3$. The rapid decay justifies differentiation, and therefore

$$\kappa = \frac{\int_0^\infty u^2\Psi(u)\,\mathrm{d}u}{2\int_0^\infty \Psi(u)\,\mathrm{d}u} > 0. \tag{10.36}$$

To see the distinction quantitatively, normalize $\mathrm{d}\nu_R = \Psi(u)\,\mathrm{d}u/\xi(1/2)$ and put $b_j = \int u^{2j}\,\mathrm{d}\nu_R$, so that $a_j = b_j/(2j)!$ and $b_0 = 1$. Then

$$m_1 = \frac{3b_1^2 - b_2}{12}. \tag{10.37}$$

Ordinary positive-measure Cauchy–Schwarz gives $b_2 \ge b_1^2$; the necessary resolvent inequality $m_1 \ge 0$ instead requires the additional upper bound $b_2 \le 3b_1^2$. These are different inequalities. The first Hankel determinant in (10.32) introduces still another combination involving b_3, and the hierarchy continues to unbounded degree.

Such kernel positivity must still be distinguished from the full resolvent criterion. For example, the entire function $\Theta_*(z) = \frac{3}{4} + \frac{1}{4}\cosh\sqrt{z}$ has a positive cosh-measure representation and is positive for $z \ge 0$. Nevertheless its normalized coefficients satisfy $a_1 = 1/8$, $a_2 = 1/96$, and its transformed moment is $m_1 = a_1^2 - 2a_2 = -1/192 < 0$. Thus even a positive cosh representation and entire square descent do not by themselves imply the desired Stieltjes moment positivity.

The exact benefit of the hyperbolic source identities is therefore the calibrated chain $I_j \longmapsto \xi^{(2n)}(1/2) \longmapsto a_n \longmapsto m_n$. The missing theorem must control these *actual transformed* moments:

for one finite $M > 0$, all finite sections of (m_{i+j}), (m_{i+j+1}), and $(M^2 m_{i+j} - m_{i+j+2})$ must be positive semidefinite. The source integrals already give positivity of ordinary log-moment matrices $(I_{i+j}(1/2))$, since for every finite coefficient list

$$\sum_{i,j} \overline{c}_i c_j I_{i+j}(1/2) = \int_0^\infty x^{-1/2} h(x) \left| \sum_j c_j (\log x)^j \right|^2 \mathrm{d}x \geq 0.$$

These are moments of $\log x$, which ranges over the whole real line. They are not already moments of a bounded nonnegative operator with the required logarithmic-derivative calibration. Transferring their positivity through (10.28) and (10.30) is a substantive additional obligation. Establishing it uniformly would activate the paper's bounded moment reconstruction and RH transfer.

10.6 From the signed hyperbolic pencil to positive determinant models

The determinant identities in the Hyperbolic Zeta project give a second construction, different from the multiplication model above. Let k, m, N, J be integers with $k, m \geq 1$ and $1 \leq N < J$, and put $h = J - N$, $x_0 = \zeta(2k+1)$ and

$$D_j(t) = \prod_{r=1}^{j} (t + r^2).$$

For an admissible affine Hankel pencil in that project, its value G at x_0 has entries

$$G_{ij} = \int_0^\infty y^{2(i+j)} \frac{D_N(y^2)^{2m}}{D_J(y^2)} \omega_k(y)\, \mathrm{d}y, \qquad 0 \leq i, j < h. \tag{10.38}$$

All these integrals converge. For $c \in \mathbb{C}^h \setminus \{0\}$,

$$c^* G c = \int_0^\infty \left| \sum_{j=0}^{h-1} c_j y^{2j} \right|^2 \frac{D_N(y^2)^{2m}}{D_J(y^2)} \omega_k(y)\, \mathrm{d}y > 0.$$

The density and rational factor are positive, and a nonzero polynomial cannot vanish on an interval. Thus G is positive definite. The affine pencil is

$$\mathcal{G}(X) = G + (X - x_0)M, \qquad \Delta(X) = \det \mathcal{G}(X), \qquad B = G^{-1/2} M G^{-1/2}. \tag{10.39}$$

Here $M = M^*$ is the coefficient of X. These identifications are recorded in the project's exact Gram and pole-diagonalization sources [16, 17].

The positive Gram matrix does not make the coefficient M positive. In the pole basis the identity is

$$M = V \operatorname{diag}(\gamma_r)_{N<r\leq J} V^T, \qquad \operatorname{sgn} \gamma_r = (-1)^{r-1}, \tag{10.40}$$

with V invertible. In fact,

$$\gamma_r = r^{2k} \frac{D_N(-r^2)^{2m-1}}{D'_{\mathrm{tail}}(-r^2)}, \qquad D_{\mathrm{tail}}(t) = \prod_{j=N+1}^{J} (t + j^2).$$

The numerator has sign $(-1)^N$ and the denominator has sign $(-1)^{r-N-1}$, proving the sign in (10.40). Consequently B is self-adjoint and invertible, but has both positive and negative eigenvalues when $h \geq 2$. Hyperbolicity of Δ means its roots are real; the Hilbert seed instead requires a nonnegative operator. These are different properties.

Proposition 10.4 (Squared-pencil identity)**.** *For any finite self-adjoint B as in (10.39) and any $c > 0$, let $K = cB^2$ and $Q_c(z) = \det(I + zK)$. Then*

$$Q_c(z) = \frac{\Delta(x_0 + i\sqrt{cz})\Delta(x_0 - i\sqrt{cz})}{\Delta(x_0)^2}. \tag{10.41}$$

The right side is a polynomial in z, independent of the choice of square root. If $\beta_1, \dots, \beta_h$ are the eigenvalues of B, then $\lambda_j = c\beta_j^2 \geq 0$ and

$$Q_c(z) = \prod_{j=1}^{h}(1 + z\lambda_j), \qquad \frac{Q_c'(a)}{Q_c(a)} = \sum_{j=1}^{h}\frac{\lambda_j}{1 + a\lambda_j} = \langle v, (I + aK)^{-1}v\rangle, \quad a \geq 0, \tag{10.42}$$

where $v = \sum_j \sqrt{\lambda_j}\, e_j$ in an orthonormal eigenbasis of K. In particular the exact differential seed algebra holds for Q_c'/Q_c.

Proof. Congruence in (10.39) gives $\Delta(x_0 + u)/\Delta(x_0) = \det(I + uB)$. The commuting factors satisfy

$$(I + i\sqrt{cz}B)(I - i\sqrt{cz}B) = I + czB^2.$$

Taking determinants proves (10.41). The spectral theorem gives the product and the scalar resolvent in (10.42). Applying the already proved identity (3.8) to this K, v gives the differential identity. For real $z \geq 0$, the numerator in (10.41) is an absolute square; for general complex z it is the product of the two displayed factors. □

The operation $B \mapsto B^2$ changes the scalar determinant. It therefore constructs positive candidates without transferring any identity with the actual Θ. The next result states a sufficient transfer condition in terms of exactly specified coefficients.

10.7 An all-order coefficient-limit theorem

Write $\Theta(z)/\Theta(0) = \sum_{r\geq 0} a_r z^r$, where $a_0 = 1$, and let $\kappa = a_1 = L(0) > 0$ as established above. Consider a sequence of finite, nonzero self-adjoint matrices B_n, for example a specified sequence of the hyperbolic slopes. Define

$$c_n = \frac{\kappa}{\operatorname{Tr}(B_n^2)}, \qquad K_n = c_n B_n^2, \qquad Q_n(z) = \det(I + zK_n) = \sum_{r\geq 0} e_{n,r} z^r. \tag{10.43}$$

Coefficients above the degree of Q_n are set equal to zero. The denominator is positive, and

$$K_n \geq 0, \qquad \operatorname{Tr} K_n = \kappa, \qquad \|K_n\| \leq \kappa, \qquad 0 \leq e_{n,r} \leq \frac{\kappa^r}{r!}. \tag{10.44}$$

The last inequality follows by expanding $(\sum_j \lambda_{n,j})^r$: the terms with r distinct indices include $r!$ copies of every summand of the elementary symmetric polynomial $e_{n,r}$.

Theorem 10.5 (Conditional transfer of hyperbolic determinant models)**.** *For the sequence (10.43), the following two conditions are equivalent:*

$$\text{for every } r \geq 0, \qquad \lim_{n\to\infty} e_{n,r} = \frac{\xi^{(2r)}(1/2)}{(2r)!\,\xi(1/2)} = a_r; \tag{10.45}$$

$$\text{for every } r \geq 0, \qquad \lim_{n\to\infty} \operatorname{Tr}(K_n^{r+1}) = \frac{(-1)^r}{r!}L^{(r)}(0) = m_r. \tag{10.46}$$

Either condition implies locally uniform convergence on $\mathbb{C}$,

$$Q_n(z) \longrightarrow \frac{\Theta(z)}{\Theta(0)}, \tag{10.47}$$

and implies RH. No assertion that the particular Hyperbolic Zeta sequence satisfies either condition is included in this theorem.

Proof. Put $p_{n,j} = \operatorname{Tr}(K_n^j)$ for $j \geq 1$. Newton's identities give

$$re_{n,r} = \sum_{j=1}^{r} (-1)^{j-1} e_{n,r-j} p_{n,j}.$$

The actual coefficients satisfy the identical finite algebraic recurrence

$$ra_r = \sum_{j=1}^{r} (-1)^{j-1} a_{r-j} m_{j-1},$$

obtained from $\Theta' = L\Theta$. For any fixed r, the first recurrence determines $e_{n,r}$ from $p_{n,1}, \ldots, p_{n,r}$ and also determines $p_{n,r}$ from $e_{n,1}, \ldots, e_{n,r}$, since $e_{n,0} = 1$. Induction on r proves the equivalence of (10.45) and (10.46).

Assume coefficient convergence. Taking limits in (10.44) gives $0 \leq a_r \leq \kappa^r / r!$. On every disk $|z| \leq R$, both coefficient series are dominated by $\sum_r (\kappa R)^r / r! = e^{\kappa R}$. Split the series at a fixed degree, use coefficient convergence for its finite head, and then let that degree tend to infinity. This proves (10.47). Quantitatively, for every integer $d \geq 0$,

$$\sup_{|z| \leq R} \left| Q_n(z) - \frac{\Theta(z)}{\Theta(0)} \right| \leq \sum_{r=0}^{d} |e_{n,r} - a_r| R^r + \frac{2e^{\kappa R} (\kappa R)^{d+1}}{(d+1)!}. \tag{10.48}$$

Every zero of Q_n is $-1/\lambda_{n,j}$ for a positive eigenvalue. Since $\lambda_{n,j} \leq \kappa$, all Q_n are zero-free on the connected domain $\mathbb{C} \setminus (-\infty, -1/\kappa]$. Their locally uniform limit equals 1 at the origin and is not identically zero. Hurwitz's theorem therefore makes this limit zero-free on the same domain. The zeros of Θ lie on the negative real axis, giving RH by Section 2.

There is also a direct route through the moment criterion. The measures $\mu_n = \sum_j \lambda_{n,j} \delta_{\lambda_{n,j}}$ have mass κ and support in $[0, \kappa]$. Hence their moments satisfy all three quadratic-form inequalities in Section 8, with $M = \kappa$. Under (10.46), taking limits in each finite quadratic form transfers those inequalities to the actual m_r. The reconstruction theorem then supplies the calibrated seed. □

The fixed Hilbert space in this limit can be displayed directly. The measures μ_n in the proof have uniformly bounded mass on the compact interval $[0, \kappa]$, so they have weakly convergent subsequences. Condition (10.46) fixes every moment of any such limit. Polynomials are dense in the continuous functions on this interval, so the limit measure is unique; call it μ. Then

$$\mathcal{H} = L^2(\mu), \qquad (Kf)(t) = tf(t), \qquad v(t) = 1 \tag{10.49}$$

satisfy $0 \leq K \leq \kappa I$ and $\|v\|^2 = \kappa$. The Taylor series of $\langle v, (I + zK)^{-1} v \rangle = \int (1 + zt)^{-1} \, \mathrm{d}\mu(t)$ has coefficients $(-1)^r m_r$, so it equals the actual L near zero. Thus the conditional limit constructs one fixed seed of the type used in the main theorem.

For the squared pencils, the first two nonautomatic trace targets are

$$\lim_{n \to \infty} \frac{\operatorname{Tr} B_n^4}{(\operatorname{Tr} B_n^2)^2} = \frac{m_1}{\kappa^2}, \qquad \lim_{n \to \infty} \frac{\operatorname{Tr} B_n^6}{(\operatorname{Tr} B_n^2)^3} = \frac{m_2}{\kappa^3}. \tag{10.50}$$

These can serve as preliminary consistency tests for a proposed choice of pencil parameters. They become an RH argument only when the all-order statement (10.46) is proved.

Example 10.6 (Trace normalization does not give calibration)**.** Take $K_n = (\kappa/n) I_n$, which has nonnegative spectrum and trace κ. Then

$$Q_n(z) = (1 + \kappa z/n)^n \longrightarrow e^{\kappa z}, \qquad \operatorname{Tr}(K_n^{r+1}) = \kappa^{r+1} n^{-r}.$$

This limit has no zeros, whereas the actual Θ has infinitely many zeros. Thus positivity, growing dimension, exact normalization, and locally uniform convergence do not by themselves identify the limit with Θ. The example is a general normalization test, not an assertion that the project's particular pencils have this limit.

The remaining Hyperbolic Zeta obligation. Choose an explicit family of admissible pencils and prove (10.45), equivalently (10.46), for every degree. The constructions and identities above establish a positive candidate family and the exact consequence of such a limit theorem. They leave that limit theorem open. The normalization uses the actual real constant κ, and the pencil data use actual zeta values. Finite matrix dimension therefore does not make this construction a finite raw checker witness. The implication proved here is convergence of this family implies RH; a converse for this particular family would require a separate completeness theorem.

11 Why a finite matrix is not the finite witness

11.1 An obstruction to exact finite-dimensional calibration

Proposition 11.1 (Finite visible spectrum is impossible)**.** *There is no exactly calibrated seed for the actual Θ whose scalar spectral measure has finite support. In particular, no finite-dimensional or finite-rank operator can supply an exact seed.*

Proof. A finitely supported spectral measure makes $F(z) = \sum_{j=1}^{q} c_j/(1+\lambda_j z)$ rational, with a possible constant term when $\lambda_j = 0$. Write $F = p/q_0$ for polynomials, where q_0 is not identically zero. Exact interval calibration implies that the entire function

$$q_0(z)\Theta'(z) - p(z)\Theta(z)$$

vanishes on an interval and therefore vanishes on all of $\mathbb{C}$. At any zero z_0 of Θ with $q_0(z_0) \neq 0$, the local ODE $\Theta' = (p/q_0)\Theta$ contradicts the finite-order zero argument of Theorem 6.1. Thus every zero of Θ would have to be among the finitely many roots of q_0. But ξ has infinitely many distinct zeros, and the two-to-one square map can identify only opposite pairs. Hence Θ has infinitely many distinct zeros, a contradiction.

If K is finite rank, its nonzero spectral values form a finite set; its kernel can contribute only the additional spectral point zero. Its scalar spectral measure is therefore finitely supported, so the same argument applies. □

This obstruction does not prohibit a finite description of an infinite-dimensional operator or a finite certificate for an infinite analytic argument. It rules out the narrower proposal that a finite positive matrix by itself can reproduce the full actual logarithmic derivative exactly on an interval.

11.2 The two missing checker theorems

The project's requested finite endpoint is

$$\big(\exists W : \mathsf{FiniteWitness},\ \mathsf{check}(W) = \mathsf{true}\big) \quad \Longleftrightarrow \quad \mathrm{RH}. \tag{11.1}$$

The witness must consist of finite raw data and the checker must terminate. The analytic seed defined in this paper is not such a witness type: it quantifies over a Hilbert space, an operator, a vector, and an exact interval identity.

A route through Theorem 1.3 would require the following separate statements for one specified checker:

$$\mathsf{check}(W) = \mathsf{true} \implies \mathsf{HSeed}_{\Theta}, \qquad \text{for every finite } W; \tag{11.2}$$

$$\mathrm{RH} \implies \exists W,\ \mathsf{check}(W) = \mathsf{true}. \tag{11.3}$$

Equation (11.2) is actual-function soundness. It must derive calibration, not receive an unverified calibration proof as a field of the purported raw witness. Equation (11.3) is finite-language completeness. Constructing a countable spectrum from all zeros under RH proves an analytic existence statement; it does not extract an accepted finite code in an independently fixed language.

Finite moment matrices, finite Gram matrices and finite truncations are legitimate components of a possible certificate design. To yield (11.2), they need a theorem extending the verified finite information to the uniform analytic assertion. The all-order moment criterion in Corollary 8.2 specifies one precise target for such an extension. No such finite reduction is established by the present note.

11.3 Finite jets cannot replace a functional identity

There is a direct mathematical reason for the distinction. Fix a real anchor $a_* > 0$ and an integer $N \geq 0$. The functions

$$L(z) \quad \text{and} \quad L(z) + \varepsilon(z - a_*)^{N+1}$$

have identical derivatives through order N at a_* for every ε, but are different analytic functions when $\varepsilon \neq 0$. Finitely many samples can likewise be preserved by adding a nonzero polynomial vanishing at all sample points. The identity theorem requires a set with an accumulation point in the analytic domain, or equivalent exact all-order germ information. A finite initial jet is neither of these.

12 Connection with the repository and proof obligations

This note was prepared against the public repository `randogelist/Riemann-Hypothesis-Final` at the source revision

`362d9f8aeaaf9479cc27ceb6b63bdb7456267990`.

The inspected source records an analytic slit-plane admission [7]:

$$\exists A \text{ holomorphic on } \mathbb{C} \setminus (-\infty, 0], \qquad \Theta' + A\Theta = 0 \text{ there.} \tag{12.1}$$

The source declaration SoloboloAdmission.slitODEAdmission_iff_RH has literal target `_root_.RiemannHypothesis`. The resolvent transfer in this paper supplies the coefficient $A = -F$ on that domain, because the smaller slit plane is contained in D_M. The sign is negative: the paper writes $\Theta' = F\Theta$, whereas the source admission writes $\Theta' + A\Theta = 0$.

The source zero-geometry declaration AngularMaster.native_RH_iff_theta_zeros_nonpositive matches Proposition 2.1, allowing nonpositive real zeros before origin nonvanishing is supplied [8]. The repository's positive-product assessment uses the normalization $\sum_n \lambda_n/(1 + a\lambda_n)$ with $\lambda_n = \gamma_n^{-2}$, consistent with (7.6) [9].

The particular legacy archive name RH_SOLOBOLO_SEED_v0_1_0.zip and the named `NativeOperatorSeed` sources were not recovered in the inspected repository tree or targeted searches. Accordingly, Definitions 1.1 and 1.2 are explicit mathematical formulations supplied by this paper. No claim is made that their fields are textually identical to an unrecovered legacy declaration.

Layer	Statement or construction	Evidence in this note
Resolvent algebra	Norm-square identity and all-parameter Gram formula	On-paper proofs, Sections 3–4
Analytic transfer	Exact normalized or differential seed implies actual RH	On-paper proofs, Sections 5–6
Converse	RH gives the positive countable diagonal seed	On-paper proof from classical factorization, Section 7
Moment route	All-order positivity with one support bound gives a seed	On-paper reconstruction, Section 8
Hyperbolic zeta	Exact model integrals, actual-jet identities, and a conditional determinant-limit transfer	On-paper proofs, Section 10; convergence to actual coefficients remains open
Legacy seed	Exact `SeedOn`/`HasSeed` implementation	Not recovered or certified here
Lean verification	Compilation and exact endpoint axiom audit	Not run; no new formal certification
Finite checker	Actual soundness, finite completeness, accepted witness	Open; not constructed here

The repository's inspected status records native build and axiom audit as `NOT_RUN`, actual calibration as `OPEN`, and finite-witness soundness and completeness as `OPEN` [10]. Publication of this paper does not change those statuses. A later formalization must connect the exact definitions, complex derivative, domains, spectral constructions and zero geometry to the actual native declarations; familiar theorem names are not a substitute for that transport.

13 Conclusion

The recovered Hilbert-vector anchor has a complete and coherent on-paper analytic formulation. Its central scalar identity is

$$\frac{\mathrm{d}}{\mathrm{d}a}\left(a\frac{\Theta'(a)}{\Theta(a)}\right) = \|(I+aK)^{-1}v\|^2,$$

where K and v are fixed, K is bounded and nonnegative self-adjoint, and the identity is exact on a positive interval. The proof proceeds through a holomorphic resolvent, an explicitly removed integration constant, and a division-free analytic ODE. The zero-order contradiction then yields the critical-line statement.

Conversely RH supplies a diagonal seed through the genus-zero product of the entire square-descended function. This gives the analytic equivalence (1.6), the Gram identity, and the normalization (7.7). It does not construct the seed without RH. The unresolved constructive task is to derive one of these exact analytic inputs from independently verifiable data; the further finite-checker equivalence requires both (11.2) and (11.3).

Section 10 gives a precise role for the Hyperbolic Zeta identities. The Bose weights construct positive bounded model resolvents and exact odd-zeta formulas; their asymptotics exclude a direct identification with the actual L. A separate positive hyperbolic-tangent integral determines

actual ξ jets, and the Newton recurrence converts those jets to the required logarithmic moments. Squaring the project's signed pencils supplies positive determinant models. Theorem 10.5 proves that calibration of all their coefficients, or all their trace moments, would construct the fixed seed and imply RH. That all-order limit theorem, and a completeness theorem for this particular family, remain open.

A Real Hilbert spaces and the holomorphic convention

The recovered anchor may be expressed on a real Hilbert space. That does not obstruct the complex-analytic proof. If H is real, form $H_{\mathbb{C}} = H \oplus iH$ with

$$\begin{aligned}\langle x + iy, u + iw\rangle_{\mathbb{C}} = (x,u)_H + (y,w)_H \\ + i\big((x,w)_H - (y,u)_H\big).\end{aligned}$$

This inner product is conjugate-linear in the first slot and linear in the second. A real bounded self-adjoint nonnegative K extends by $K_{\mathbb{C}}(x+iy) = Kx + iKy$. Its norm is unchanged, because

$$\|K_{\mathbb{C}}(x+iy)\|^2 = \|Kx\|^2 + \|Ky\|^2 \le \|K\|^2(\|x\|^2 + \|y\|^2),$$

and restriction to real vectors gives the reverse norm inequality. Self-adjointness follows by expanding the inner product. Positivity follows from

$$\langle x + iy, K_{\mathbb{C}}(x+iy)\rangle = (x, Kx)_H + (y, Ky)_H \ge 0,$$

since the imaginary cross terms cancel by symmetry. For real $a \ge 0$, the inverse extends the real inverse, so every real scalar calibration and norm identity is preserved. Thus the complex theorem applies to the real Hilbert version as well.

If an alternative convention makes the inner product linear in the first slot, the scalar holomorphic expression should instead be $\langle R_z v, v\rangle$. With the convention fixed in this paper, it is $\langle v, R_z v\rangle$. Mixing these conventions would turn a holomorphic function into an anti-holomorphic one and invalidate the identity-theorem step.

B A compact proof dependency ledger

The forward implication has the following dependencies, each proved or cited at its point of use:

- **F1.** The actual Θ is entire and $\Theta(0) > 0$.
- **F2.** Fixed bounded positive K gives a holomorphic resolvent on D_M, a connected domain containing zero and the positive ray.
- **F3.** Exact interval calibration gives $\Theta' = F\Theta$ by the identity theorem. For differential calibration, first remove C using $Q(0) = -C\Theta(0)$.
- **F4.** A nonzero entire solution of that ODE has no zero where its coefficient is holomorphic, by the finite-order contradiction.
- **F5.** Negative-real-axis confinement of Θ zeros is equivalent to the critical-line condition for actual ξ zeros.

Only the converse uses RH:

- **R1.** RH places the zeros of Θ at $-\gamma_n^2$.
- **R2.** The order bound and Hadamard factorization give $\Theta/\Theta(0) = \prod_n (1 + z/\gamma_n^2)$ and summability.

R3. Uniformly convergent logarithmic differentiation gives $L(a) = \sum_n (a + \gamma_n^2)^{-1}$.

R4. The explicit diagonal K and square-summable v realize this scalar function and hence both seed conditions.

The all-order moment criterion reconstructs the forward input through Theorem 8.1. It does not remove an infinite quantifier by itself. Neither direction in this ledger uses a finite accepted witness, and neither is reported as a Lean-verified endpoint.

References

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[2] NIST Digital Library of Mathematical Functions, *Section 25.2: Definition and Expansions*, especially Eq. 25.2.3. https://dlmf.nist.gov/25.2. Accessed 27 October 2026.

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[7] Riemann-Hypothesis-Final project, `SoloboloAdmission/NativeODE.lean`, source revision `362d9f8`. Exact source of `slitODEAdmission_iff_RH`; native audit not run in this note. Pinned repository source.

[8] Riemann-Hypothesis-Final project, `AngularMaster/ZeroGeometry.lean`, source revision `362d9f8`. Pinned repository source.

[9] Riemann-Hypothesis-Final project, *RH positive-product assessment*, 25 October 2026, source revision `362d9f8`. Pinned research assessment.

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[12] zetahyperbolic-axiom project, `Theorem_3_1_Hyperbolic_Weight.lean`, revision `b56c810`, declarations `density` and `density_pos`. Source inspection; no fresh native audit. Pinned weight source.

[13] zetahyperbolic-axiom project, `Theorem_3_1_Hyperbolic_Moments.lean`, revision `b56c810`, declaration `integral_monomial_density`. Pinned moment source.

[14] zetahyperbolic-axiom project, `Theorem_3_1_Pole_Integral_Identity.lean`, revision `b56c810`, declaration `poleIntegralIdentity`. Its integer-parameter statement is extended on paper to real positive parameters in Proposition 10.2. Pinned pole-integral source.

[15] zetahyperbolic-axiom project, `LeanPaper/ContinuousDensityNormalization.lean`, revision `b56c810`, declarations `correctedDensity_odd` and `printedDensity_odd`. Pinned normalization correction.

[16] zetahyperbolic-axiom project, `LeanPaper/ContinuousGram.lean`, revision `b56c810`, declarations `baseMatrix_entry_exact`, `baseMatrix_posDef`, and `pencil_odd`. Pinned Gram source.

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[22] NIST Digital Library of Mathematical Functions, *Section 5.5: Functional Relations*, Eq. 5.5.5 (gamma duplication). https://dlmf.nist.gov/5.5. Accessed 27 October 2026.

[23] George Csordas, *Fourier transforms of positive definite kernels and the Riemann ξ-function*, arXiv:1309.0055v2, 21 February 2014, Section 4, Eqs. (4.1)–(4.2), printed pp. 9–10. The Fourier-variable ξ in that source is distinguished from the completion $\xi(s)$ used here by the rescaling in (10.35). https://arxiv.org/abs/1309.0055.